Non-existence of Negative Weight Derivations of the Moduli Algebras of Isolated Weighted Homogeneous Hypersurface Singularities

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Abstract

Mather and Yau proved that the analytic isomorphism class of an isolated complex hypersurface singularity is determined by its dimension and its moduli algebra. Yau's conjecture asserts that the moduli algebra of an isolated weighted homogeneous hypersurface singularity over $\mathbb{C}$ has no nonzero derivation of negative degree. We prove this conjecture.

1 Introduction

Let $f \in \mathbb{C}\{x_1, \ldots, x_n\}$ satisfy $f(0) = 0$, and suppose that 0 is an isolated critical point of f. We regard $(V(f), 0)$ as the analytic hypersurface germ with local ring $\mathbb{C}\{x_1, \ldots, x_n\}/(f)$. Thus all first partial derivatives of f vanish at 0, and they have no other common zero in a sufficiently small neighborhood. Let $S = \mathbb{C}[x_1, \ldots, x_n]$, $\mathfrak{m} = (x_1, \ldots, x_n)$, $\mathcal{O} = \mathbb{C}\{x_1, \ldots, x_n\}$. The Milnor algebra and the Tjurina algebra (also called moduli algebra) of f are

$$M_f = \mathcal{O}/(\partial f/\partial x_1, \ldots, \partial f/\partial x_n), \qquad T_f = \mathcal{O}/(f, \partial f/\partial x_1, \ldots, \partial f/\partial x_n).$$

Both algebras are finite dimensional over $\mathbb{C}$.

Mather-Yau Theorem ([6])**.** *Let $f, g \in \mathbb{C}\{x_1, \ldots, x_n\}$ define isolated hypersurface singularities at 0 of same dimension. Then the analytic germs $(V(f), 0)$ and $(V(g), 0)$ are isomorphic if and only if T_f and T_g are isomorphic as $\mathbb{C}$-algebras.*

Assume from now on that f is weighted homogeneous. This means that there are positive rational numbers $w_1, \ldots, w_n, d$ such that every monomial $x_1^{\alpha_1} \cdots x_n^{\alpha_n}$ with nonzero coefficient in f satisfies $\sum_i \alpha_i w_i = d$. The weights are positive, so only finitely many such monomials exist and f is a polynomial. Multiplying all weights and d by a common positive integer makes them integers. We use integer weights throughout.

Give x_i weight w_i. For $a \in \mathbb{Z}$, let S_a be the vector space spanned by the monomials of weight a. Then $S_a = 0$ for $a < 0$, $S_0 = \mathbb{C}$, and $S = \bigoplus_{a \geq 0} S_a$. An element of S_a is homogeneous of weighted degree a. Weighted degree need not agree with ordinary total degree. Unless ordinary degree is specified, degree means weighted degree.

We use $\partial_i = \partial/\partial x_i$, $f_i = \partial_i f$, $J = (f_1, \ldots, f_n) \subset S$, and $A = S/J$. The ideal J is the Jacobian ideal. Lemma A.1 identifies A with the analytic Milnor algebra M_f.

Euler's identity gives $df = \sum_i w_i x_i f_i$. It follows that $T_f = M_f$. These algebras inherit the grading defined by the weights. For a graded $\mathbb{C}$-algebra $A = \bigoplus_{j \geq 0} A_j$, a derivation is a $\mathbb{C}$-linear map $\delta : A \to A$ with

$$\delta(ab) = \delta(a)b + a\delta(b) \qquad (a, b \in A).$$

It has degree k if $\delta(A_j) \subseteq A_{j+k}$ for every j. We set $A_j = 0$ when $j < 0$, and denote the space of degree-k derivations by $\mathrm{Der}_{\mathbb{C}}(A)_k$. Lemma B.3 gives each homogeneous derivation

δ of negative degree on A a unique homogeneous polynomial lift $D = \sum_i h_i \partial_i$ that preserves J.

Yau's conjecture. *If f is weighted homogeneous and has an isolated critical point at 0, then $\mathrm{Der}_{\mathbb{C}}(M_f)_k = 0$ for every $k < 0$.*

The conjecture arose from Yau's study of derivations of moduli algebras. It is related to the microlocal characterization of quasihomogeneous singularities developed by Xu and Yau [11]. Chen, Xu and Yau attribute the conjecture to Yau's work on that characterization [3, p. 244].

Chen, Xu and Yau proved the cases of two and three variables [3, Theorems 2.2 and 2.3]. Chen proved the case of four variables [2, Theorem 2.1]. Yau and Zuo proved the conjecture in any number of variables under the weight conditions $d \geq 2w_1 \geq \cdots \geq 2w_n > 0$ and $w_1 \leq 2w_n$ [12, Main Theorem, p. 21]. Chen, Yau and Zuo proved the conjecture for weighted homogeneous fewnomial singularities of multiplicity at least five [5, Main Theorem C]. In that result, a fewnomial singularity admits a defining polynomial with exactly n monomials in n variables [5, Definition 9.1], and its multiplicity is the least ordinary total degree of a monomial with nonzero coefficient. Chen and Yau later obtained the sufficient bound $d > W^2 - W + 1$, where W is the largest positive integer weight [4, Theorem 1.5].

We prove the conjecture in any number of variables.

Main Theorem. *Let $f \in \mathbb{C}\{x_1, \ldots, x_n\}$ be weighted homogeneous with positive weights and have an isolated critical point at 0. Then*

$$\mathrm{Der}_{\mathbb{C}}(M_f)_k = 0 \qquad \textit{for every } k < 0.$$

We briefly describe the proof.

Step 1: Splitting off the quadratic variables by weight-preserving coordinate changes reduces the problem to $A = \mathbb{C}$ or to $f \in \mathfrak{m}^3$ with $d > 2 \max_i w_i$. In the latter case, if A has a nonzero derivation of negative degree, Proposition 2.4 produces a nonzero homogeneous polynomial derivation D of negative degree such that $D(J) \subseteq J$ and $D^2 f = 0$. This specially chosen D is locally nilpotent.

Step 2: For $r > 0$, consider the ideal generated by the partial derivatives in variables of weight greater than r. We prove that its radical contains no nonzero polynomial of weight r. For small r, we use a square deformation and Lemma 3.4, which derives the required first-order consequence from [1, Theorem 14.8]. For larger r, the result follows from the quadratic dependence on the variables of weight greater than r.

Step 3: Lemma 4.1 forces D to vanish on the reduced quotient by a suitable ideal of partial derivatives. Choose a variable of least weight moved by D. Its image under D must then lie in the radical just described, in the degree excluded by the radical result. This is the contradiction.

2 Removal of quadratic variables and the choice of D

A coordinate change in this section means a weight-preserving polynomial automorphism of S. It is therefore a graded automorphism. By the chain rule, its invertible Jacobian matrix maps one gradient ideal to the other. It therefore induces a graded isomorphism of Milnor algebras.

Lemma 2.1. *Let f be a weighted homogeneous polynomial with an isolated critical point at 0. After weight-preserving coordinate changes and removal of the quadratic variables, the graded algebra $A = S/J$ is isomorphic either to $\mathbb{C}$ or to the Milnor algebra of a weighted homogeneous polynomial, again denoted by f, satisfying*

$$f \in \mathfrak{m}^3, \qquad d > 2W, \qquad W = \max_i w_i.$$

The remaining polynomial has an isolated critical point.

Proof. First suppose that a variable u has maximal weight $W > d/2$. No monomial of f contains u twice. Write the other variables as z. Then

$$f = uL(z) + F(z), \qquad \deg L = d - W.$$

We have $L(0) = 0$, since the origin is critical. The polynomial L must have a nonzero linear term. Otherwise all derivatives of f vanish on the u-axis: the u derivative is $L(0) = 0$, and each z derivative is $uL_{z_i}(0) + F_{z_i}(0) = 0$. That would contradict isolatedness.

Choose a variable v occurring linearly in L. Its weight is $d-W < W$. After a linear change among variables of weight $d - W$, followed by a nonzero scalar change, write $L = v + P(z')$, where z' denotes all other variables. Every nonlinear monomial of weight $d - W$ uses only variables of smaller weight. Thus P does not involve v, and the replacement $v' = L$ is a weight-preserving polynomial automorphism. Rename v' as v. Now $f = uv + F(v, z')$.

The polynomial $F(v, z') - F(0, z')$ is divisible by v. Put

$$K(v, z') = \frac{F(v, z') - F(0, z')}{v}, \qquad u' = u + K(v, z').$$

The degree of K is W. It is independent of u. Hence this is another weight-preserving polynomial automorphism, and

$$f = u'v + F(0, z').$$

Its gradient ideal contains u' and v. Its graded Milnor algebra is exactly the graded Milnor algebra of $F(0, z')$. The critical set is the product of the origin in the (u', v) variables with the critical set of $F(0, z')$. Thus the latter critical point is isolated. Repeat this operation. Each step removes two variables, so it stops. Every remaining weight is at most $d/2$.

If some remaining variables have weight $d/2$, collect them in a column vector u and write the other variables as z. The weighted degree condition gives

$$f = \tfrac{1}{2}u^{\mathsf{T}}Hu + u^{\mathsf{T}}a(z) + b(z),$$

where H is a constant symmetric matrix. Each entry of a has weight $d/2$. Since every z variable has smaller weight, a has no constant or linear terms.

The matrix H is invertible. If $0 \neq v \in \ker H$, then $(u, z) = (tv, 0)$ is critical for every t. Indeed, the u derivatives are $Hu + a(0) = 0$, and the z derivatives vanish because a has no linear terms and $db(0) = 0$. This would again contradict isolatedness.

Set $u' = u + H^{-1}a(z)$. This preserves the weights and gives

$$f = \tfrac{1}{2}(u')^{\mathsf{T}}Hu' + b(z) - \tfrac{1}{2}a(z)^{\mathsf{T}}H^{-1}a(z).$$

The gradient ideal of the first summand is the ideal generated by the entries of u'. Removing these variables therefore preserves the graded Milnor algebra and isolatedness.

All remaining weights are strictly less than $d/2$. Hence no monomial of ordinary degree two has weight d. There are also no constant or linear terms. The remaining polynomial belongs to $\mathfrak{m}^3$ and satisfies $d > 2W$. If no variables remain, the quotient algebra is $\mathbb{C}$. $\square$

We reuse the symbols x_i, w_i, f, S, J, A for the remaining variables, weights, polynomial, and algebras after splitting. From now on, unless otherwise stated, $f \in \mathfrak{m}^3$ and $d > 2W$. In particular, every f_i has weighted degree at least $d - W > W$.

Define the graded vector space

$$\mathcal{V} = \{g \in J : \partial_i g \in J \text{ for every } i\}, \qquad \mathcal{V}_q = \mathcal{V} \cap S_q.$$

Lemma 2.2. *For an integer q with $0 < q < d$, the map*

$$\begin{aligned} \mathrm{Der}_{\mathbb{C}}(A)_{q-d} &\longrightarrow \mathcal{V}_q, \\ \delta &\longmapsto Df, \end{aligned}$$

is a linear bijection, where D is the unique homogeneous polynomial lift of δ given by Lemma B.3.

Proof. Let $\delta \in \mathrm{Der}_{\mathbb{C}}(A)_{q-d}$ and write its unique lift as $D = \sum_j h_j \partial_j$. Lemma B.3 gives $D(J) \subseteq J$. Then $Df = \sum_j h_j f_j \in J$. The product rule gives, for every i,

$$\partial_i(Df) = D(f_i) + \sum_j (\partial_i h_j) f_j \in J.$$

Thus $Df \in \mathcal{V}_q$.

Conversely, let $g \in \mathcal{V}_q$. Since $g \in J$, taking homogeneous components gives $g = \sum_j h_j f_j$ with $h_j \in S_{q-d+w_j}$. Define $D = \sum_j h_j \partial_j$. The same identity gives

$$D(f_i) = \partial_i g - \sum_j (\partial_i h_j) f_j \in J.$$

Thus D preserves J and induces a derivation $\delta \in \mathrm{Der}_{\mathbb{C}}(A)_{q-d}$. Lemma B.2 excludes nonzero constant coefficients. The other coefficients have positive degree, so all belong to $\mathfrak{m}$. They are the unique lifts in Lemma B.3. Lemma B.1 proves uniqueness of the expression for g, and hence injectivity of the map. □

Lemma 2.3. *If D is any polynomial derivation with $D(J) \subseteq J$, then $D(\mathcal{V}) \subseteq \mathcal{V}$.*

Proof. For $g \in \mathcal{V}$, we have $Dg \in J$. With $D = \sum_j h_j \partial_j$, differentiation gives

$$\partial_i(Dg) = D(\partial_i g) + \sum_j (\partial_i h_j)(\partial_j g).$$

The first term belongs to J because $\partial_i g \in J$ and D preserves J. Every term in the sum belongs to J because $\partial_j g \in J$. Hence $Dg \in \mathcal{V}$. □

Proposition 2.4. *If A has a nonzero homogeneous derivation of negative degree, then there is a nonzero homogeneous polynomial derivation D of degree $-s$, with $s > 0$, such that*

$$D(J) \subseteq J, \qquad 0 \neq g = Df, \qquad Dg = D^2 f = 0.$$

Every coefficient of D belongs to $\mathfrak{m}$, and D is locally nilpotent.

Proof. Let $0 \neq \delta \in \mathrm{Der}_{\mathbb{C}}(A)_{-s_0}$ with $s_0 > 0$, and let D_0 be its lift from Lemma B.3. The lift is nonzero, and $D_0 f \neq 0$ by Lemma B.1, since its total weighted degree is $d - s_0 < d < 2(d - W)$. The product rule gives $D_0 f \in \mathcal{V}$. As J has no nonzero element of nonpositive degree, $0 < d - s_0 < d$. Hence $\mathcal{V}_q \neq 0$ for some integer q with $0 < q < d$.

Choose the least such integer q and a nonzero $g \in \mathcal{V}_q$. Lemma 2.2 gives a derivation D with $Df = g$ and degree $-s = q - d < 0$. Lemma 2.3 gives $Dg \in \mathcal{V}$. Its degree is $q - s < q$. If that degree is positive, minimality of q implies $Dg = 0$. If it is nonpositive, membership in J implies the same conclusion. The remaining assertions follow from Lemma B.3. □

3 Radicals of ideals generated by selected partial derivatives

Lemma 3.1. *Let $F(z,Y)$ be a convergent power series germ (or polynomial), where $z = (z_1,\dots,z_p)$ and $Y = (Y_1,\dots,Y_q)$, with $q \geq 1$. Suppose that F has an isolated critical point at 0, belongs to $(z,Y)^3$, and is polynomial of ordinary degree at most two in Y. Write*

$$F(z,Y) = \tfrac{1}{2}Y^{\mathsf{T}}H(z)Y + L(z)^{\mathsf{T}}Y + c(z), \qquad H(z)^{\mathsf{T}} = H(z).$$

Then $\det H(z)$ is not the zero power series (or zero polynomial). If $k(z)$ is a column of q convergent power series (or polynomials) and $k(z)^{\mathsf{T}}F_Y(z,Y)$ is independent of Y, then $k = 0$. Here F_Y is the column of the Y derivatives.

Proof. The assumption $F \in (z,Y)^3$ gives $H(0) = 0$, $L \in (z)^2$, and $c \in (z)^3$. Put $H_j = (\partial H/\partial z_j)(0)$. Along $z = 0$, the derivatives are

$$F_Y(0,Y) = 0, \qquad F_{z_j}(0,Y) = \tfrac{1}{2}Y^{\mathsf{T}}H_jY.$$

The only common zero of the quadratic forms $Y^{\mathsf{T}}H_jY$ is $Y = 0$. Otherwise, if there exists a nonzero vector v such that $v^TH_jv = 0$ for every j. Then every point $(z,Y) = (0,tv)$, for sufficiently small t, would be a critical point, contradicting the assumption that the origin is an isolated critical point. In particular, the case $p = 0$ is impossible under the hypotheses.

Let $\mathcal{H}$ be the vector space spanned by the symmetric matrices $H_1,\dots,H_p$ over $\mathbb{C}$. We show that it contains an invertible matrix. Choose $M \in \mathcal{H}$ of maximal rank e. If $e < q$, choose $0 \neq v \in \ker M$. The symmetric bilinear form with matrix M has an e-dimensional subspace U on which it is nondegenerate. This follows by choosing a basis for its nonsingular part; equivalently, a symmetric bilinear form in characteristic different from two can be diagonalized by a change of basis. Since v is in the kernel, $U \cap \mathbb{C}v = 0$.

Take any $N \in \mathcal{H}$. The rank of $M + tN$ is at most e for every $t \in \mathbb{C}$, by maximality of e. Its restriction to $U \oplus \mathbb{C}v$ therefore has determinant zero for every t. In a basis consisting of a basis of U followed by v, that restriction has matrix

$$\begin{pmatrix} M_U + tN_U & tb \\ tb^{\mathsf{T}} & tv^{\mathsf{T}}Nv \end{pmatrix}.$$

The coefficient of t in its determinant is $\det(M_U)v^{\mathsf{T}}Nv$. Terms that use an off-diagonal entry in the last row must also use one in the last column, so they have order at least two in t. Since M_U is invertible, we obtain $v^{\mathsf{T}}Nv = 0$. If $e = 0$, the same argument uses the one-dimensional space $\mathbb{C}v$ and gives this equality directly.

This holds for every $N \in \mathcal{H}$, hence for every H_j. It contradicts the absence of a common nonzero zero of the quadratic forms. Thus $e = q$, and $\mathcal{H}$ contains an invertible matrix.

The linear part of $H(z)$ is $\sum_j z_jH_j$. Its determinant is a nonzero homogeneous polynomial of ordinary degree q. Because every entry of $H(0)$ is zero, this polynomial is exactly the ordinary degree-q part of $\det H(z)$. Hence $\det H$ is not identically zero.

Finally, $F_Y = H(z)Y + L(z)$. The coefficient of Y in $k(z)^{\mathsf{T}}F_Y$ is $k(z)^{\mathsf{T}}H(z)$. Independence of Y implies $H(z)k(z) = 0$. Over the fraction field of $\mathbb{C}\{z\}$ the matrix H is invertible, so $k = 0$. For polynomials, the same argument uses $\mathbb{C}(z)$, the fraction field of $\mathbb{C}[z]$. □

Lemma 3.2. *Assume $f \in S_d \cap \mathfrak{m}^3$, $d > 2W$, and f has an isolated critical point. If $0 \neq a \in S_r$ and $0 < 3r < d$, then $a^2 \notin J$.*

Proof. Suppose instead that $a^2 \in J$. Write

$$a^2 = \sum_i k_i f_i, \qquad k_i \in S_{2r-d+w_i}.$$

Only indices with $w_i \geq d - 2r$ can have nonzero coefficients. Denote all variables with these weights by Y, and all other variables by z. If there are no Y variables, all k_i vanish, which already contradicts $a \neq 0$.

Since $3r < d$, every Y variable has weight at least $d - 2r > r$ and also greater than $d/3$. Thus a is independent of Y. Further,

$$\deg k_i \leq W + 2r - d < r.$$

The strict inequality follows from $W < d/2$ and $r < d/3$, which imply $W + r < d$. Therefore every nonzero k_i is also independent of Y.

Three Y variables, counted with repetition, have total weight greater than d. Hence f has ordinary degree at most two in Y. The displayed expression for a^2 now has the form

$$k(z)^{\mathsf{T}} f_Y(z, Y) = a(z)^2.$$

Its right side is independent of Y. Lemma 3.1 forces $k = 0$. This gives $a^2 = 0$ in the polynomial ring, contrary to $a \neq 0$. □

For an analytic germ F with isolated critical point, its Milnor number is

$$\mu(F) = \dim_{\mathbb{C}} \mathcal{O}/(F_{x_1}, \ldots, F_{x_n}).$$

A one-parameter analytic deformation of f is a convergent power series $F(x, t)$ with $F(x, 0) = f(x)$. We write $f_t(x) = F(x, t)$. It has constant Milnor number at the origin if 0 is critical for all small t, remains isolated, and $\mu(f_t) = \mu(f)$.

Lemma 3.3. *Let $f(u, y)$ be an analytic germ with an isolated critical point at 0 and $f(0) = 0$. Let $u = (u_1, \ldots, u_p)$ with $p \geq 1$, and $y = (y_1, \ldots, y_q)$. Suppose that $a = a(u)$ belongs to the radical of $(f_{y_1}, \ldots, f_{y_q})$. Then $f_t = f + ta^2$ has an isolated critical point at 0 and $\mu(f_t) = \mu(f)$ for each $t \in \mathbb{C}$, as a germ at 0. For the analytic family near $t = 0$, the critical set near $(0, 0)$ is, set-theoretically, the zero section $x = 0$, and its critical value is zero.*

Proof. Write $\mathcal{O} = \mathbb{C}\{u, y\}$ and

$$R = \mathcal{O}/(f_{y_1}, \ldots, f_{y_q}).$$

The complete gradient is a regular sequence in $\mathcal{O}$, in any order. Hence R is Cohen–Macaulay of dimension p. The assumption on a means that its image in R is nilpotent. In particular $a(0) = 0$, since the ideal of the y derivatives vanishes at the origin.

Use bars for classes in R and set

$$Q_t = (\overline{f_{u_i}} + 2t\overline{a}\,\overline{a_{u_i}} : 1 \leq i \leq p) \subset R.$$

The image of $\overline{a}$ is zero in R_{red}. Thus Q_t and Q_0 have the same image in R_{red} and the same radical in R. The quotient R/Q_0 is the Milnor algebra of f, so it has finite length. The same is true of R/Q_t. Each Q_t is therefore a parameter ideal in the p-dimensional ring R.

Let $\mathfrak{p}$ be a minimal prime with $\dim R/\mathfrak{p} = p$. A nilpotent element belongs to every prime ideal, so $\overline{a} \in \mathfrak{p}$. It follows that the images of Q_t and Q_0 in $R/\mathfrak{p}$ are equal. The equality on

each minimal-prime quotient allows us to apply the associativity formula. Since the ideals are parameter ideals in a Cohen–Macaulay ring, their quotient lengths equal their multiplicities. Hence,

$$\begin{aligned}\operatorname{length}_R(R/Q_t) &= e(Q_t, R)\\ &= \sum_{\substack{\mathfrak{p}\in\operatorname{Min} R\\ \dim R/\mathfrak{p}=p}} \operatorname{length}_{R_\mathfrak{p}}(R_\mathfrak{p})\, e(Q_t(R/\mathfrak{p}), R/\mathfrak{p})\\ &= \sum_{\substack{\mathfrak{p}\in\operatorname{Min} R\\ \dim R/\mathfrak{p}=p}} \operatorname{length}_{R_\mathfrak{p}}(R_\mathfrak{p})\, e(Q_0(R/\mathfrak{p}), R/\mathfrak{p})\\ &= e(Q_0, R) = \operatorname{length}_R(R/Q_0).\end{aligned}$$

The y derivatives of f_t are those of f, since a depends only on u. The u derivatives are $f_{u_i} + 2taa_{u_i}$. Therefore R/Q_t is the Milnor algebra of f_t. The displayed equality proves constancy of the Milnor number. Since $a(0) = 0$, all first derivatives of f_t vanish at 0. The finite length just proved gives isolatedness.

We also check the critical set of the family. Work in $\mathbb{C}\{u, y, t\}/(f_y)$. The relation making a nilpotent still holds in this ring. After reduction, the ideals generated by $f_u + 2taa_u$ and by f_u agree. The full gradient ideal of f_t and the extension of the full gradient ideal of f thus have the same radical in $\mathbb{C}\{u, y, t\}$. The latter radical is (u, y). Hence, set-theoretically, the critical set of the family near $(0, 0)$ is exactly $u = y = 0$. Finally $f_t(0) = f(0) + ta(0)^2 = 0$, so the critical value stays zero. □

The algebraic argument uses only the first-order consequence stated below for a linear one-parameter family. We derive it from the description of the reduced μ-constant locus in [1], Theorem 14.8. The stratum theorem is due to Varchenko [9, Theorem 2, p. 30]. Saito [7, Section 3.4, pp. 160–161] gives another statement of the same stratum description. The assumptions $f \in \mathfrak{m}^3$ and $d > 2W$ are not required for the next lemma.

Lemma 3.4. *Let $f \in S_d$ have an isolated critical point at the origin and satisfy $f(0) = 0$. Let $q \in S_e$, where $e < d$. Suppose that, for every sufficiently small $t \in \mathbb{C}$, the germ $f_t = f + tq$ has an isolated critical point at the origin, $f_t(0) = 0$, and $\mu(f_t, 0) = \mu(f, 0)$. Then $q \in J$.*

Proof. The assertion is immediate if $q = 0$. First suppose that $n = 1$. Write $f = cx^m$ with $c \neq 0$ and $m \geq 2$. If $q \neq 0$, homogeneity gives $q = bx^\ell$ with $b \neq 0$ and $0 \leq \ell < m$. The condition $f_t(0) = 0$ excludes $\ell = 0$, and criticality at the origin excludes $\ell = 1$. For $2 \leq \ell < m$ and $t \neq 0$,

$$f_t'(x) = x^{\ell-1}\big(mc\,x^{m-\ell} + t\ell b\big).$$

The expression in parentheses is a unit at the origin, so $\mu(f_t, 0) = \ell - 1 < m - 1 = \mu(f, 0)$, a contradiction. Thus $q = 0$ in the one-variable case.

Now assume $n > 1$. We use the description of the μ-constant stratum in [1, Theorem 14.8]. The construction and versality of the deformation below are given in [1, Section 3.1, pp. 67–68]. Choose homogeneous monomial representatives $\phi_0 = 1, \phi_1, \ldots, \phi_{\mu-1}$ of a basis of M_f, where $\mu = \mu(f, 0)$, and put $e_j = \deg \phi_j$. The miniversal deformation for right equivalence of functions is

$$F_s(x) = f(x) + \sum_{j=0}^{\mu-1} s_j \phi_j(x).$$

Normalize the variable weights to w_i/d, so that f has degree one. The corresponding parameter weights are

$$\nu_j = 1 - \frac{e_j}{d}, \qquad \nu_j > 0 \iff e_j < d.$$

Let Σ_μ be the reduced analytic germ of parameters s for which F_s has a critical point p near the origin satisfying $F_s(p) = 0$ and $\mu(F_s, p) = \mu$. Theorem 14.8 in [1] gives

$$\Sigma_\mu = \{s : s_j = 0 \text{ whenever } \nu_j > 0\}.$$

Analytic versality gives an analytic map $\gamma(t)$ with $\gamma(0) = 0$ and a local biholomorphism $(x, t) \mapsto (\Phi(x, t), t)$ with $\Phi(x, 0) = x$, such that

$$f + tq = F_{\gamma(t)} \circ \Phi_t, \qquad \Phi_t(x) = \Phi(x, t).$$

This is right equivalence of functions. Put $p(t) = \Phi_t(0)$. The chain rule and the invertibility of the differential of Φ_t give $dF_{\gamma(t)}(p(t)) = 0$. Moreover,

$$F_{\gamma(t)}(p(t)) = f_t(0) = 0, \qquad \mu(F_{\gamma(t)}, p(t)) = \mu(f_t, 0) = \mu.$$

Consequently $\gamma(t) \in \Sigma_\mu$, so $\gamma_j(t) = 0$ whenever $e_j < d$. Differentiating the analytic identity at $t = 0$ and writing $v_i(x) = \partial_t \Phi_i(x, 0)$ gives

$$q = \sum_i v_i f_i + \sum_{e_j \geq d} \gamma_j'(0) \phi_j.$$

Take its weighted homogeneous component of degree e. The second sum has no such component because $e < d$. Hence

$$q = \sum_i (v_i)_{e-d+w_i} f_i \in J.$$

Each coefficient $(v_i)_{e-d+w_i}$ is a polynomial because all variable weights are positive; a component of negative degree is zero. □

Corollary 3.5. *Assume $f \in S_d \cap \mathfrak{m}^3$, $d > 2W$, and that the origin is an isolated critical point. Let $a \in S_r$, where $r > 0$ and $3r < d$. If*

$$a \in \sqrt{(f_i : w_i > r)},$$

then $a^2 \in J$.

Proof. If $a = 0$, the assertion is immediate. Assume $a \neq 0$. Let u denote the variables of weight at most r, and y the remaining variables. Positivity of the weights implies that $a = a(u)$. The list u is nonempty, since a nonzero polynomial of positive degree r cannot be formed using only variables of weight greater than r. If the list y is empty, the ideal in the hypothesis is zero, whose radical is zero in S, contrary to $a \neq 0$.

The radical relation remains valid in the analytic local ring, because the same identity $a^N \in (f_y)$ holds there for some $N \geq 1$. Lemma 3.3 therefore applies to $f + ta^2$. It gives an isolated critical point at the origin, constant local Milnor number, and critical value zero for all sufficiently small t. Since $a^2 \in S_{2r}$ and $2r < d$, Lemma 3.4 gives $a^2 \in J$. The stronger bound $3r < d$ is retained because it is also needed in Lemma 3.2. □

For a positive integer r, define

$$I_r = (f_i : w_i > r) \subset S.$$

The ideal generated by an empty list is the zero ideal.

Theorem 3.6. *Let $f \in S_d \cap \mathfrak{m}^3$ have an isolated critical point, and assume $d > 2W$. Then*

$$\sqrt{I_r} \cap S_r = \{0\} \qquad \text{for every integer } r > 0.$$

Proof. Suppose that $0 \neq a \in \sqrt{I_r} \cap S_r$. Let u be the list of variables of weight at most r, and let y be the list of variables of weight greater than r. Since a has degree r, it only depends on u. If there are no u variables, then $S_r = 0$, a contradiction. If there are no y variables, then $I_r = 0$, whose radical is zero in S, again a contradiction. We can therefore assume both lists are nonempty. We have $I_r = (f_y)$.

First suppose that $3r < d$. Corollary 3.5 gives $a^2 \in J$. This contradicts Lemma 3.2.

It remains to treat $3r \geq d$. Every y variable has weight strictly greater than $r \geq d/3$. A product of three y variables has weight greater than d. Thus f has ordinary degree at most two in y. Write

$$f(u, y) = \tfrac{1}{2} y^{\mathsf{T}} H(u) y + L(u)^{\mathsf{T}} y + c(u).$$

By Lemma 3.1, $\det H(u)$ is nonzero. Let $K = \mathbb{C}(u)$, the fraction field of the polynomial ring in the u variables. Over K the matrix H is invertible, so the equations $f_y = Hy + L$ are equivalent to $y = -H^{-1}L$. It follows that

$$K[y]/(f_y) \simeq K.$$

In particular, (f_y) is a proper ideal of $K[y]$.

On the other hand, $a \in \sqrt{I_r}$ means that $a^N \in (f_y)$ for some $N \geq 1$. Since a is a nonzero polynomial in u, it becomes a unit in K. After extension to $K[y]$, the ideal (f_y) would contain a unit. This contradicts its being proper. Both ranges of r are excluded. □

4 Proof of the main theorem

A locally nilpotent derivation δ of a $\mathbb{C}$-algebra R gives automorphisms

$$\exp(t\delta)(a) = \sum_{\nu \geq 0} \frac{t^\nu}{\nu!} \delta^\nu(a) \qquad (a \in R).$$

The sum is finite for each a. The product rule implies that this map is an algebra homomorphism. Its inverse is $\exp(-t\delta)$, and the maps satisfy the group law for $(\mathbb{C}, +)$. They give the associated additive-group action.

For a finitely generated $\mathbb{C}$-algebra R with generators $z_1, \ldots, z_N$, we regard a complex point as a $\mathbb{C}$-algebra homomorphism $p : R \to \mathbb{C}$. Its coordinates are $z_j(p) = p(z_j)$. These points are exactly the tuples in $\mathbb{C}^N$ satisfying all polynomial relations among the generators. An element of R vanishes at every complex point if and only if it is nilpotent, by the Nullstellensatz. This point description is all that we need from the affine spectrum $\operatorname{Spec} R$.

Lemma 4.1. *Let $R = \bigoplus_{a \geq 0} R_a$ be a finitely generated graded $\mathbb{C}$-algebra with $R_0 = \mathbb{C}$. Suppose that homogeneous elements $z_1, \ldots, z_N$ generate it, with positive integer degrees $w_j \leq W$. Let $q_1, \ldots, q_m$ be homogeneous elements of degrees $d_i > W$ such that $R/(q_1, \ldots, q_m)$ is finite dimensional. If δ is a locally nilpotent derivation of R and*

$$\delta^2 q_i = 0 \qquad (1 \leq i \leq m),$$

then the derivation induced by δ on R_{red} is zero. The derivation δ is not required to be homogeneous.

Proof. First, δ preserves the nilradical. For a nilpotent element a, each automorphism $\exp(t\delta)$ sends a to a nilpotent element. Thus, in R_{red}, the polynomial

$$\sum_{\nu \geq 0} \frac{t^\nu}{\nu!}[\delta^\nu a]$$

is zero for every $t \in \mathbb{C}$. Evaluate it at more distinct scalars than its degree. The invertibility of the resulting Vandermonde matrix shows that each coefficient is zero. In particular δa is nilpotent. Hence δ induces a derivation on R_{red}.

Assume that this induced derivation is nonzero. Since the classes of the z_j generate R_{red}, at least one δz_j is nonzero there. The Nullstellensatz gives a complex point p at which this element does not vanish. The orbit through p has coordinate polynomials

$$P_j(t) = (\exp(t\delta)z_j)(p).$$

At least one P_j is nonconstant because $P_j'(0) = (\delta z_j)(p)$ is nonzero for some j.

For every nonconstant P_j, let $N_j \geq 1$ be its ordinary degree in t. Define

$$\alpha = \min_{P_j \text{ nonconstant}} \frac{w_j}{N_j} = \frac{a}{b},$$

where a, b are positive integers. We have $0 < \alpha \leq W$. Introduce a new parameter ζ and set

$$\gamma_j(\zeta) = \zeta^{bw_j} P_j(\zeta^{-a}).$$

Each γ_j is a polynomial in ζ. Indeed, if a term of P_j has degree $k \leq N_j$, its exponent after substitution is $bw_j - ak \geq bw_j - aN_j \geq 0$. For a constant P_j, the exponent is $bw_j > 0$. A zero polynomial causes no difficulty.

Choose an index attaining the minimum defining α. The leading term of its P_j gives exponent zero, and all lower terms give positive exponents. Its leading coefficient is nonzero. Therefore $\gamma_j(0) \neq 0$ for this index.

The grading defines the scaling action $z_j \mapsto \lambda^{w_j} z_j$. For $\zeta \neq 0$, the tuple $\gamma(\zeta)$ is obtained by first taking the additive orbit at $t = \zeta^{-a}$, and then scaling with $\lambda = \zeta^b$. It is therefore a point of $\operatorname{Spec} R$. Every defining polynomial relation vanishes on $\gamma(\zeta)$ for nonzero ζ. Since each coordinate is polynomial, the relation also vanishes at $\zeta = 0$. Thus $\gamma(0)$ is a point of $\operatorname{Spec} R$.

Because $\delta^2 q_i = 0$, the orbit of q_i is affine in t:

$$(\exp(t\delta)q_i)(p) = q_i(p) + t(\delta q_i)(p).$$

Homogeneity now gives, for $\zeta \neq 0$,

$$q_i(\gamma(\zeta)) = \zeta^{bd_i} q_i(p) + \zeta^{bd_i - a}(\delta q_i)(p).$$

Both exponents are positive. For the second, use $d_i > W \geq \alpha = a/b$. Hence $q_i(\gamma(0)) = 0$ for every i.

The only common zero of the q_i is the point where all z_j vanish. To check this, note that the quotient by the q_i is finite dimensional and graded. It has zero homogeneous components in all sufficiently large degrees. A sufficiently high power of each positive-degree z_j is therefore zero in that quotient. Thus every common zero of the q_i has all $z_j = 0$.

This contradicts the nonzero coordinate of $\gamma(0)$ found above. The induced derivation on R_{red} must be zero. □

Remark 4.2. Zero elements among the q_i may be omitted. The argument does not require R to be reduced, a domain, or Cohen–Macaulay. It also does not require the number of q_i to equal $\dim R$.

Lemma 4.3. *Let $D = \sum_j h_j \partial_j$ be homogeneous of degree $-s < 0$. Assume that $D(J) \subseteq J$ and $D^2 f = 0$. Fix a positive integer r such that $h_j = 0$ whenever $w_j \leq r$. Then D preserves I_r, and its induced derivation on $(S/I_r)_{\mathrm{red}}$ is zero.*

Proof. For each i, the polynomial $D(f_i)$ belongs to J and has degree $d - w_i - s$. Taking homogeneous coefficients gives

$$D(f_i) = \sum_j c_{ij} f_j, \qquad c_{ij} \in S_{w_j - w_i - s}.$$

If $w_i > r$ and $c_{ij} \neq 0$, then $w_j - w_i - s \geq 0$, so $w_j \geq w_i + s > r$. Every term belongs to I_r. This proves $D(I_r) \subseteq I_r$.

Put $g = Df$ and $g_i = \partial_i g$. By the product rule and the hypothesis on the coefficients h_j,

$$g_i = D(f_i) + \sum_{w_j > r} (\partial_i h_j) f_j.$$

Therefore $D(f_i) \equiv g_i \pmod{I_r}$ for every i. When $w_i > r$, the already proved stability of I_r also gives $g_i \in I_r$.

The assumption $D^2 f = 0$ says $Dg = 0$. Differentiate this equality with respect to x_i:

$$0 = \partial_i(Dg) = D(g_i) + \sum_{w_j > r} (\partial_i h_j) g_j.$$

For every index in the sum, $g_j \in I_r$. Consequently $D(g_i) \in I_r$ for all i. Apply D to $D(f_i) - g_i \in I_r$. Since $D(I_r) \subseteq I_r$, we obtain

$$D^2(f_i) - D(g_i) \in I_r, \qquad \text{hence } D^2(f_i) \in I_r.$$

Now let $R = S/I_r$. It is graded, has $R_0 = \mathbb{C}$, and is generated by the classes of the variables, of degrees at most W. The negative degree of D makes it locally nilpotent, by the degree argument in Lemma B.3. Its induced derivation on R is also locally nilpotent. For the indices with $w_i \leq r$, take q_i to be the image of f_i in R. Each nonzero such element is homogeneous of degree

$$d - w_i \geq d - W > W.$$

The remaining partial derivatives are already zero in R. Thus

$$R/(q_i : w_i \leq r) = S/J,$$

which is finite dimensional. The calculation above gives $D^2 q_i = 0$. Lemma 4.1 applies and shows that the induced derivation on R_{red} is zero. □

Proof of the Main Theorem. Multiply the positive rational weights by a common positive integer. This changes no sign of a derivation degree. The function is a polynomial because all weights are positive. Lemma A.1 identifies its analytic Milnor algebra with the polynomial one. Euler's identity identifies this algebra with the moduli algebra.

By Lemma 2.1, a graded algebra isomorphism carries degree-k derivations to degree-k derivations by conjugation. Thus it suffices to prove the assertion for the algebra after

splitting. If that algebra is $\mathbb{C}$, every $\mathbb{C}$-linear derivation is zero. Otherwise we have $f \in \mathfrak{m}^3$ and $d > 2W$.

Suppose, for a contradiction, that there is a nonzero homogeneous derivation of negative degree. Proposition 2.4 gives a nonzero polynomial derivation D of degree $-s < 0$ such that

$$D(J) \subseteq J, \qquad D^2 f = 0, \qquad h_i = D(x_i) \in \mathfrak{m} \cap S_{w_i - s}.$$

Choose an index k for which $h_k \neq 0$ and w_k is as small as possible. Set

$$h = h_k, \qquad r = w_k - s.$$

The polynomial h is nonzero, homogeneous, and belongs to $\mathfrak{m}$. Hence its degree r is a positive integer. Since $s > 0$, we also have $r < w_k$.

If $w_i \leq r$, then $w_i < w_k$. By the choice of k, $h_i = 0$. The hypotheses of Lemma 4.3 are therefore satisfied. It follows that the induced derivation on $(S/I_r)_{\mathrm{red}}$ is zero.

In particular, the class of $D(x_k) = h$ is zero in $(S/I_r)_{\mathrm{red}}$. By the definition of reduction, this means that the image of h in S/I_r is nilpotent. Equivalently, $h^N \in I_r$ for some $N \geq 1$, so

$$0 \neq h \in \sqrt{I_r} \cap S_r.$$

This contradicts Theorem 3.6. Thus no nonzero homogeneous negative derivation exists. Undoing the graded isomorphisms from the splitting gives the stated conclusion for the original Milnor and moduli algebras. □

Remark 4.4. For a finite-dimensional graded algebra, every derivation is a finite sum of homogeneous derivations. Indeed, for homogeneous $a \in A_j$, project $\delta(a)$ onto A_{j+k} to define its degree-k component. Projecting the product rule shows that each component is a derivation. Only finitely many components occur because only finitely many A_j are nonzero. The Main Theorem therefore says that the entire derivation algebra has no negative graded part.

A Local algebras

In this appendix, we recall some basic facts about local algebras. All rings in this paper are commutative with identity. A ring is Noetherian if every ideal is finitely generated. A local ring has one maximal ideal. Its residue field is the quotient by that ideal. Its dimension is its Krull dimension: the supremum of the lengths of strict chains of prime ideals. An ideal is prime if its quotient ring is an integral domain. A prime is minimal if it is minimal among all prime ideals of the ring. We write $\operatorname{Min} R$ for this set.

Let $(R, \mathfrak{n})$ be a Noetherian local ring. A sequence $a_1, \ldots, a_c \in \mathfrak{n}$ is regular if multiplication by a_i is injective on $R/(a_1, \ldots, a_{i-1})$ for each i. A local ring is Cohen–Macaulay if it has a regular sequence in its maximal ideal of length $\dim R$. A system of parameters in an r-dimensional local ring is a list of r elements of its maximal ideal whose generated ideal has finite-length quotient. That ideal is called a parameter ideal. The length $\operatorname{length}_R M$ of a module M is the number of simple factors in a composition series. If R is a local $\mathbb{C}$-algebra with residue field $\mathbb{C}$, a finite-length module has length equal to its dimension over $\mathbb{C}$. This applies to $\mathcal{O}$ and its quotients, but not in general to their localizations at other prime ideals.

We use the following standard facts. The local rings $S_{\mathfrak{m}}$ and $\mathcal{O}$ have dimension n and are Cohen–Macaulay. Here $S_{\mathfrak{m}}$ is obtained by allowing denominators outside $\mathfrak{m}$. In a Cohen–Macaulay local ring every system of parameters is a regular sequence, in any order. A

quotient by c members of a regular sequence is Cohen–Macaulay of dimension lower by c. These statements follow, for example, from [8, Tags 00NQ and 02JN].

Since J has finite colength, $f_1, \dots, f_n$ is a regular sequence in $S_{\mathfrak{m}}$ and in $\mathcal{O}$. It is also regular in S. To see the last assertion, let H be any homogeneous ideal generated by an initial part of this sequence. The map $S/H \to (S/H)_{\mathfrak{m}}$ is injective. Indeed, if $ub = 0$ in S/H for a polynomial u with $u(0) \neq 0$ and a nonzero element b, the component of least degree of ub is $u(0)$ times the component of least degree of b. It cannot vanish. A failure of regularity in S would therefore remain a failure after localization, which is impossible. The same argument applies after any reordering.

If Q is a proper ideal with finite-length quotient in an r-dimensional local ring, its Hilbert–Samuel multiplicity $e(Q, R)$ is $r!$ times the coefficient of N^r in the polynomial that equals $\operatorname{length}_R(R/Q^N)$ for all sufficiently large integers N. If R is Cohen–Macaulay and Q is a parameter ideal, then

$$\operatorname{length}_R(R/Q) = e(Q, R).$$

Here is one way to obtain the length formula. For a regular sequence generating Q, the standard description of the associated graded ring is

$$\operatorname{gr}_Q R := \bigoplus_{j \geq 0} Q^j/Q^{j+1} \simeq (R/Q)[T_1, \dots, T_r],$$

where T_i is the degree-one class of the ith generator; see [8, Tag 00LN]. There are $\binom{j+r-1}{r-1}$ monomials of degree j in r variables when $r > 0$. Summing the lengths of Q^j/Q^{j+1} for $0 \leq j < N$ gives

$$\operatorname{length}_R(R/Q^N) = \binom{N+r-1}{r} \operatorname{length}_R(R/Q).$$

Comparison of leading coefficients gives the length formula. If $r = 0$, $Q = (0)$ and the formula follows directly from the definition.

The associativity formula expresses the same multiplicity as

$$e(Q, R) = \sum_{\substack{\mathfrak{p} \in \operatorname{Min} R \\ \dim R/\mathfrak{p} = r}} \operatorname{length}_{R_{\mathfrak{p}}}(R_{\mathfrak{p}})\, e(Q(R/\mathfrak{p}), R/\mathfrak{p}).$$

The notation $Q(R/\mathfrak{p})$ means the image of Q in $R/\mathfrak{p}$. The ring $R_{\mathfrak{p}}$ is the localization at $\mathfrak{p}$. The associativity formula is [8, Tag 0AZX]. The length formula also follows from [8, Tag 0AZZ], because the higher Koszul homology groups of a regular sequence vanish. We will apply them only to quotients of $\mathcal{O}$ by part of the gradient sequence.

For an ideal I in a ring R, its radical is

$$\sqrt{I} = \{a \in R : a^N \in I \text{ for some integer } N \geq 1\}.$$

An element is nilpotent if some positive power is zero. The ideal $\sqrt{(0)}$ is the nilradical of R. The reduced quotient is $R_{\mathrm{red}} = R/\sqrt{(0)}$. A ring is reduced if its nilradical is zero.

Lemma A.1. *If the weighted homogeneous polynomial f has an isolated critical point at 0, then $\sqrt{J} = \mathfrak{m}$. Moreover, the natural map*

$$S/J \longrightarrow \mathcal{O}/J\mathcal{O}$$

is an isomorphism. In particular, A is a finite-dimensional local algebra with maximal ideal $\mathfrak{m}/J$.

Proof. For $\lambda \in \mathbb{C}^*$, define $\lambda \cdot x = (\lambda^{w_1} x_1, \ldots, \lambda^{w_n} x_n)$. Homogeneity gives

$$f_i(\lambda \cdot x) = \lambda^{d-w_i} f_i(x).$$

If a nonzero point p were critical, then every $\lambda \cdot p$ would be critical. As λ tends to zero, these nonzero points tend to zero. This contradicts the isolated critical point assumption. Thus the common zero set of J in $\mathbb{C}^n$ is $\{0\}$. The Hilbert Nullstellensatz gives $\sqrt{J} = \mathfrak{m}$. The form of the Nullstellensatz used here says that a polynomial vanishing on the common zeros of an ideal has some power in that ideal.

For each i, choose N_i with $x_i^{N_i} \in J$. If $N = 1 + \sum_i (N_i - 1)$, every monomial of ordinary degree at least N is divisible by some $x_i^{N_i}$. Hence $\mathfrak{m}^N \subset J$. Modulo $J\mathcal{O}$, a convergent power series equals its Taylor polynomial of ordinary degree less than N. This proves surjectivity of the displayed map.

To prove injectivity, suppose that $p \in S$ and $p = \sum_i b_i f_i$ with $b_i \in \mathcal{O}$. Expand each b_i into its weighted homogeneous components. Each such component is a polynomial, since all weights are positive. Taking the component of degree a gives

$$p_a = \sum_i (b_i)_{a-d+w_i} f_i \in J.$$

The polynomial p has finitely many components. Their sum belongs to J. Thus the map is injective. The quotient is finite dimensional because $\mathfrak{m}^N \subset J$. It is local because its only maximal ideal is $\mathfrak{m}/J$. □

We will also use the argument in the last paragraph for an ideal generated by any list of homogeneous polynomials. In particular, analytic membership of a homogeneous polynomial in such an ideal implies polynomial membership.

B Locally nilpotent derivations

Throughout this appendix, $f \in S_d \cap \mathfrak{m}^3$ has an isolated critical point at the origin and $d > 2W$, where $W = \max_i w_i$, as obtained after the reduction in Lemma 2.1.

Lemma B.1. *Suppose that $q < 2(d-W)$, that $a_i \in S_{q-d+w_i}$ for each i, and that $\sum_i a_i f_i = 0$. Then all coefficients a_i are zero. Consequently, an expression $g = \sum_i a_i f_i$ with g homogeneous of degree less than d has unique homogeneous coefficients of degrees $\deg g - d + w_i$.*

Proof. Suppose some a_i is nonzero. Let j be the largest such index. If $j = 1$, then $a_1 f_1 = 0$ contradicts regularity. If $j > 1$, the relation implies

$$a_j f_j \in (f_1, \ldots, f_{j-1}).$$

Since f_j is a nonzero divisor modulo this ideal, a_j belongs to $(f_1, \ldots, f_{j-1})$. This homogeneous ideal has no nonzero element of degree less than $d - W$. Hence $\deg a_j \geq d - W$. Since $\deg f_j \geq d - W$, the total degree is at least $2(d - W)$, a contradiction.

For uniqueness, subtract two homogeneous expressions for g. Their difference is a relation of total degree $\deg g < d < 2(d - W)$. The first assertion applies. □

Lemma B.2. *Every derivation of a finite-dimensional local $\mathbb{C}$-algebra preserves its maximal ideal.*

Proof. Let B be such an algebra and let $\mathfrak{n}$ be its maximal ideal. Each element of $\mathfrak{n}$ is nilpotent. This follows, for example, from the nilpotence of the maximal ideal in an Artinian local ring. It also follows directly by applying stabilization of the ideals $\mathfrak{n}^j$ and Nakayama's lemma. Nakayama's lemma says that a finite module M with $M = \mathfrak{n}M$ is zero.

Take $0 \neq a \in \mathfrak{n}$, and choose the least integer N with $a^N = 0$. If $\delta(a) \notin \mathfrak{n}$, it is a unit. Applying δ gives

$$0 = \delta(a^N) = Na^{N-1}\delta(a).$$

Multiplication by the inverse of $N\delta(a)$ would give $a^{N-1} = 0$, contrary to the choice of N. Therefore $\delta(a) \in \mathfrak{n}$. The assertion for $a = 0$ is immediate. □

Lemma B.3. *Let s be a positive integer. Every $\delta \in \mathrm{Der}_{\mathbb{C}}(A)_{-s}$ has a unique lift*

$$D = \sum_{i=1}^{n} h_i \frac{\partial}{\partial x_i}, \qquad h_i \in \mathfrak{m} \cap S_{w_i - s}, \qquad D(J) \subseteq J.$$

The derivation D is locally nilpotent: for every $p \in S$, some power $D^N(p)$ is zero.

Proof. The element $\delta([x_i])$ has degree $w_i - s < W$, where brackets denote the class in A. The ideal J has no nonzero element in that degree. Hence $\delta([x_i])$ has a unique homogeneous representative $h_i \in S_{w_i - s}$. If this degree is negative, the representative is zero. If it is zero, Lemma B.2 shows that the representative is zero as well. In all cases $h_i \in \mathfrak{m}$.

Define D by the displayed formula, and let $\pi : S \to A$ be the quotient map. The maps πD and $\delta\pi$ agree on the generators. The product rule then shows that they agree on all of S. If $p \in J$, this gives $\pi(Dp) = \delta(\pi p) = 0$. Hence $D(J) \subseteq J$, and D induces δ on the quotient. Uniqueness follows from uniqueness of each h_i.

For a homogeneous polynomial p of degree a, the polynomial $D^N(p)$ has degree $a - Ns$, or is zero. It is zero when $Ns > a$. Every polynomial is a finite sum of homogeneous polynomials, so one sufficiently large N works for the whole polynomial. □

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